

Common Power Format

CPF 1.1 Pocket Guide



What is CPF?

CPF (Common Power Format) is a design specification language that addresses the limitation in traditional design automation tool flows by capturing the designer's intent for power management and enabling the automation of advanced power-lowering design techniques. The Common Power Format enables all design, verification, implementation - and technology-related power objectives to be captured in a single file and allows the application of that data across the design flow, providing a consistent reference point for design development and production.

CPF is available at Si2's OpenEDA.si2.org web site at no cost for everyone to download and use. Download instructions, along with an FAQ with important details on this process, as well as more information on the LPC is located at this link: <http://www.si2.org/?page=811>.

Available supporting tools include a parser, the CPF Pocket Guide - an easy-to-use ready reference book, and the CPF Relational Analyzer - a powerful training and analysis tool (available to LPC members). In addition, training presentations in English, Mandarin, and Japanese are also available for the CPF user.

What is the Low Power Coalition?

The Low-Power Coalition (LPC) was formed in late 2006 to deliver enhanced capabilities in low-power Integrated Circuit (IC) design flows relating to specifications of low-power design intent, architectural tradeoffs, logical/physical implementation, design verification and testability. The intent of the LPC is to develop a unified system approach to low power chip design covering the entire design flow, incorporating all applicable standards as necessary. The LPC is also planning a roadmap of additional enhancements and optimizations for reusable IP blocks and upstream migration of low-power intent.

Low Power Coalition Members

Advanced Micro Devices	Freescale
ARM	Global UniChip Corp.
Atrenta	IBM Corporation
Azuro, Inc	Intel Corporation
Cadence Design Systems	LSI
Calypto Design Systems	NXP Semiconductors
ChipVision Design Systems	Sequence Design, Inc.
Entasys Design	Virage Logic
Envis Corp.	

The Low Power Coalition at Si2 guides the evolution of the Si2 Common Power Format Standard as well as other design flow-related standards and supporting material.

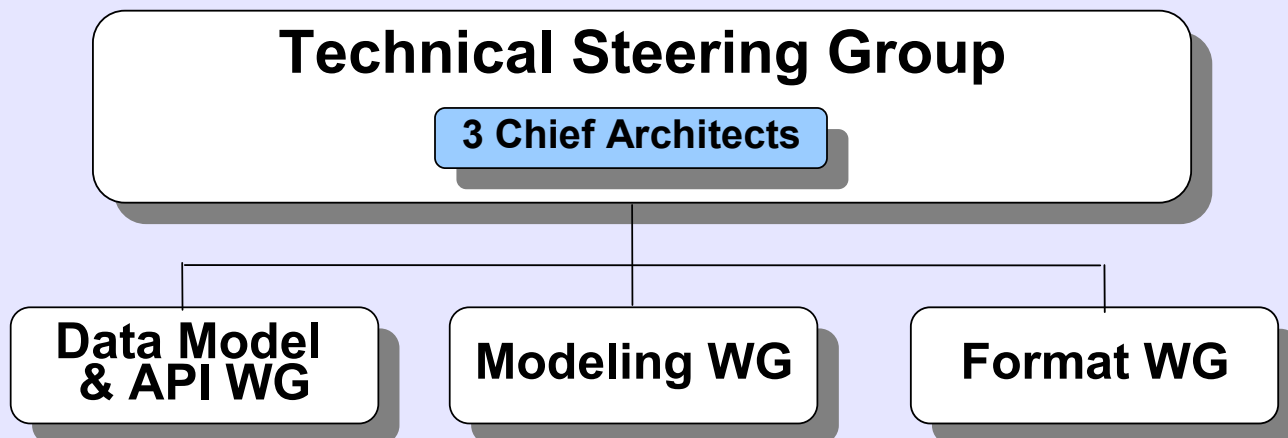
All companies are welcome to join in this unique innovation in the EDA industry.

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LPC Structure

Full LPC Membership



LPC: business/policy & standards approvals (AMD, Chair)

- **Technical Steering Group (TSG):** charts working groups, owns the low-power technology roadmap

Includes 3 Chief Architects (Cadence, IBM, LSI)

- **Active and completed working groups:**

Data Model and API WG – map clear semantics and data relationships in CPF, add API interface support to CPF

ModelingWG – define requirements and attributes of power models needed at all levels of a power-aware design flow

Format WG – define priorities and detailed requirements for all revisions of CPF

Format Comparison WG – report on technical comparison of CPF and UPF (Done, results widely shared)

Who Supports CPF?

Industry Leaders

AMD	LSI
Freescale	NEC
Fujitsu	NXP
IBM	STARC
Intel	Wipro

Foundry Reference Flows

Common Platform
SMIC
TSMC
UMC

IP Vendors

ARC	MIPS
ARM	Tensilica
Chipidea	Virage Logic
Denali	

EDA Vendors

ARC	Calypto
Apache	Chipvision
Atrenta	Entasys
Azuro	Envis
Cadence	Sequence

Design Services

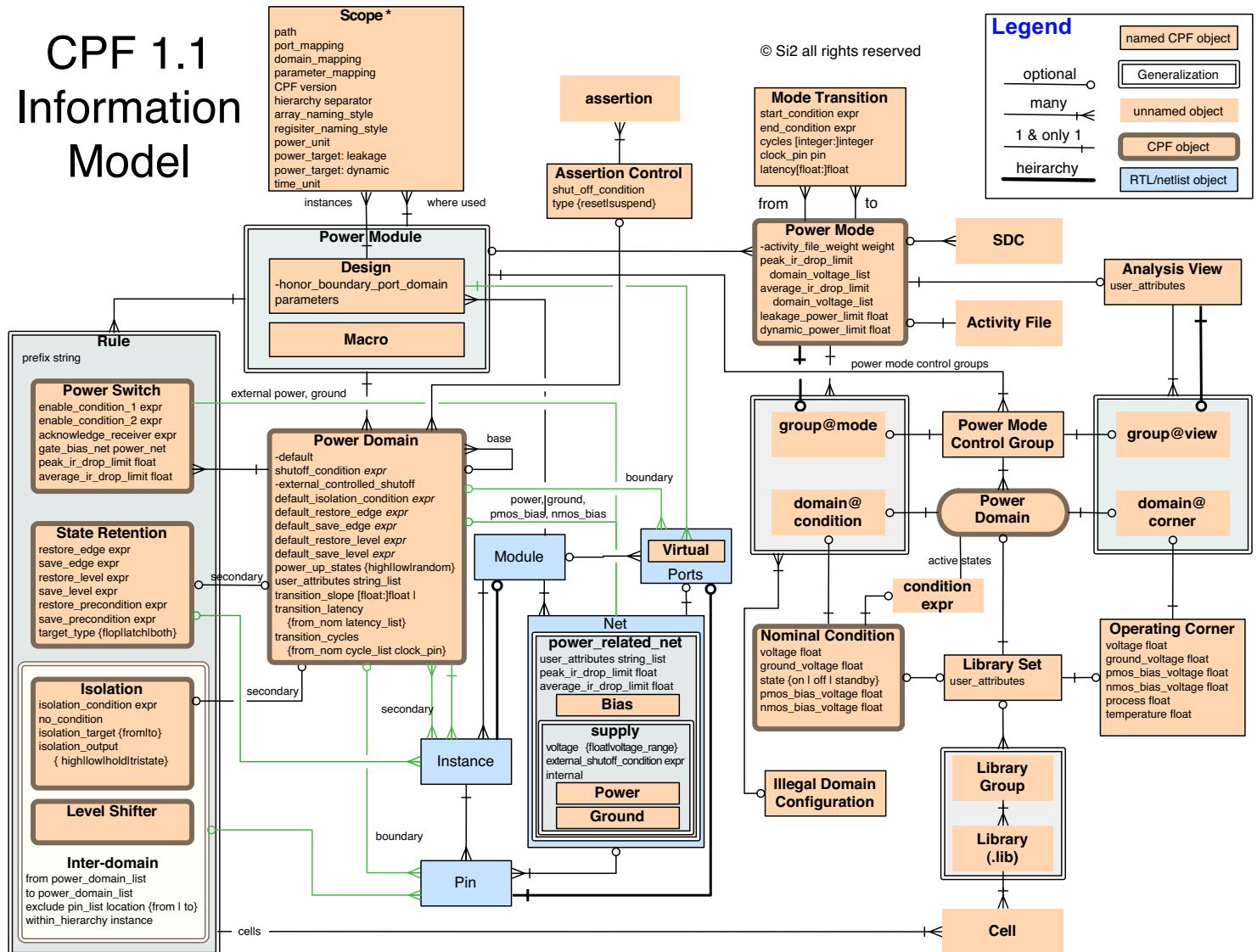
Alchip	NEC
DNP	NSW
Faraday	SOCLE
Fujitsu	Toppan
Global Unichip	VeriSilicon
MindTree	

CPF Momentum

- **Over 100 CPF design flow engagements now active**
- **Completed tape-outs include:**
 - NEC (several), Fujitsu (several), Freescale (several), NXP (several), TSMC (several), Global UniChip (several), Faraday (several), AMD
- **10 EDA tool suppliers committed to CPF (with more coming)**
 - ARC, Apache, Atrenta, Azuro, Cadence, Calypto, Chipvision, Entasys Design, Envis, Sequence Design

CPF Information Model

CPF 1.1 Information Model



This CPF information model shows how the constraints in CPF interact with a design.

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The syntax and grammar of the CPF commands in this guide are described with Extended Backus Naur form (EBNF).

Extendend Backus Naur Form	Examples
A vertical bar <code> </code> separates a list of alternative items. Only one item from the list is used. <pre>set_power_unit { pW nW uW mW W }</pre>	<pre>set_power_unit pW set_power_unit nW set_power_unit uW set_power_unit mW set_power_unit W error ← set_power_unit MW since MW (Megawatt) is not an list item</pre>
Square brackets <code>[]</code> indicate than an argument is optional. Everything enclosed between the two brackets is not used if that option is not used. <pre>set_time_unit [ns us ms]</pre>	<pre>set_time_unit ns set_time_unit us set_time_unit ms current_unit ← set_time_unit no argument returns the current time unit</pre>
Curly braces <code>}</code> enclose a hierarchical lists of items <pre>create_power_domain -name power_domain { { -default [-instances instance_list] [-boundary_ports pin_list] } { -instances instance_list [-boundary_ports pin_list] } { -instances instance_list } }</pre>	<pre>create_power_domain -name PD1 -default create_power_domain -name PD1 -default / -instances {inst_A inst_B} create_power_domain -name PD2 / -instances {inst_A inst_B}</pre>
Curly braces <code>{ }+</code> indicate that one or more of the previous list can be used <pre>create_level_shifter_rule -name string {-pins pin_list -from power_domain_list -to power_domain_list } +</pre>	<pre>create_level_shifter_rule -name shiftUp / -pins {adder[0] adder[1]} -from {PD1 PD2} / -pins {adder[15] adder[16]} -from {PD1 PD2} / -pins {adder[30] adder[31]} -from {PD1 PD2}</pre>
The data type of each argument is specified <pre>set_switching_activity { -all -pins pin_list [-instances instance_list [-hierarchical]] { -probability float -toggle_rate float [-clock_pins pin_list] -toggle_percentage float } [-mode mode]</pre>	<pre>set_switching_activity -pins {enable[1] enable[2]} \ -probability 0.65 -toggle_rate 300000124.5</pre>

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CPF Command Syntax	Description and Examples
<pre>assert_illegal_domain_configurations -name <i>string</i> { -domain_conditions <i>domain_condition_list</i> -group_modes <i>group_modes_list</i> -domain_conditions <i>domain_condition_list</i> -group_modes <i>group_mode_list</i> }</pre>	Asserts that a particular configuration of domain conditions and/or power mode control group conditions is illegal.
<pre>create_analysis_view -name <i>string</i> -mode <i>mode</i> [-user_attributes <i>string_list</i>] { -domain_corners <i>domain_corner_list</i> -group_views <i>group_view_list</i> -domain_corners <i>domain_corner_list</i> -group_views <i>group_view_list</i> }</pre>	Creates an analysis view and associates a list of operating corners with a given mode.
<pre>create_assertion_control -name <i>string</i> { -assertions <i>assertion_list</i> -domains <i>power_domain_list</i> } [-exclude <i>assertion_list</i>] [-shutoff_condition <i>expression</i>] [-type {reset suspend}]</pre>	Inhibits evaluation of any selected assertion instance when its related power domain is powered down.
<pre>create_bias_net -net <i>net</i> [-driver <i>pin</i>] [-user_attributes <i>string_list</i>] [-peak_ir_drop_limit <i>float</i>] [-average_ir_drop_limit <i>float</i>]</pre>	Specifies or creates a bias net to be used as a power supply to either forward or backward bias a transistor.
<pre>create_global_connection -net <i>net</i> -pins <i>pin_list</i> [-domain <i>domain</i> -instances <i>instance_list</i>]</pre>	Specifies how to connect a global net to the specified pins. A global net can be a data net, bias net, power net or ground net.
<pre>create_ground_nets -nets <i>net_list</i> [-voltage {<i>float</i> <i>voltage_range</i>}] [-external_shutoff_condition <i>expression</i> -internal] [-user_attributes <i>string_list</i>] [-peak_ir_drop_limit <i>float</i>] [-average_ir_drop_limit <i>float</i>]</pre>	Specifies or creates a list of ground nets.
<pre>create_isolation_rule -name <i>string</i> [-isolation_condition <i>expression</i> -no_condition] {-pins <i>pin_list</i> -from <i>power_domain_list</i> -to <i>power_domain_list</i>}+ [-exclude <i>pin_list</i>] [-isolation_target {from to}] [-isolation_output { high low hold tristate}] [-secondary_domain <i>power_domain</i>]</pre>	Defines a rule for adding isolation cells and specify which net segments must be isolated.

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<pre>create_level_shifter_rule -name string {-pins pin_list -from power_domain_list -to power_domain_list}+ [-exclude pin_list]</pre>	Defines a rule for adding level shifters.
<pre>create_mode_transition -name string -from power_mode -to power_mode -start_condition expression [-end_condition expression] [-cycles [integer:]integer -clock_pin clock_pin -latency [float:]float]</pre>	Describes how the transition between two power modes is controlled, and the time it takes for each power domain to complete the transition.
<pre>create_nominal_condition -name string -voltage float [-ground_voltage float] [-state {on off standby}] [-pmos_bias_voltage float] [-nmos_bias_voltage float]</pre>	Creates a nominal operating condition with the specified voltage
<pre>create_operating_corner -name string -voltage float [-ground_voltage float] [-pmos_bias_voltage float] [-nmos_bias_voltage float] [-process float] [-temperature float] -library_set library_set</pre>	Defines an operating corner and associate it with a library set.
<pre>create_power_domain -name power_domain [-instances instance_list] [-boundary_ports pin_list] [-default] [-shutoff_condition expression] [-external_controlled_shutoff]] [-default_isolation_condition expression] [-default_restore_edge expr -default_save_edge expr -default_restore_edge expr -default_save_edge expr -default_restore_level expr -default_save_level expr] [-power_up_states {high low random}] [-active_state_conditions active_state_condition_list] [-base_domains domain_list]</pre>	Creates a power domain and specifies the instances, boundary ports and pins that belong to this power domain.
<pre>create_power_mode -name string [-default] {-domain_conditions domain_condition_list -group_modes group_mode_list -domain_conditions domain_condition_list -group_modes group_mode_list }</pre>	Defines a power mode. One and only one power mode can be the default power mode.

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<pre>create_power_nets -nets <i>net_list</i> [-voltage {<i>float</i> <i>voltage_range</i>}] [-external_shutoff_condition <i>expression</i> -internal] [-user_attributes <i>string_list</i>] [-peak_ir_drop_limit <i>float</i>] [-average_ir_drop_limit <i>float</i>]</pre>	Specifies or creates a list of power nets. The power nets are created within the current scope.
<pre>create_power_switch_rule -name <i>string</i> -domain <i>power_domain</i> {-external_power_net <i>net</i> -external_ground_net <i>net</i>}</pre>	Specifies how a single on-chip power switch must connect the external power or ground nets to the primary power or ground nets of the specified power domain.
<pre>create_state_retention_rule -name <i>string</i> { -domain <i>power_domain</i> -instances <i>instance_list</i> } [-exclude <i>instance_list</i>] [-restore_edge <i>expr</i> -save_edge <i>expr</i> -restore_edge <i>expr</i> -save_edge <i>expr</i> -restore_level <i>expr</i> -save_level <i>expr</i>] [-restore_precondition <i>expr</i>] [-save_precondition <i>expr</i>] [-target_type {<i>flop</i> <i>latch</i> <i>both</i>}] [-secondary_domain <i>domain</i>]</pre>	Defines a rule for replacing either the selected registers in the instance list or all of the registers in the specified power domain with state retention registers.
<pre>define_library_set -name <i>library_set</i> -libraries <i>list</i> [-user_attributes <i>string_list</i>]</pre>	Creates a library set.
<pre>end_design [<i>module</i>]</pre>	Used with a <code>set_design</code> command, groups a number of CPF commands that apply to the current design or top design.
<pre>end_macro_model [<i>macro_cell</i>]</pre>	Used with a <code>set_macro_model</code> command, groups a number of CPF commands that apply to the described macro cell.
<pre>end_power_mode_control_group</pre>	Used with a <code>set_power_mode_control_group</code> command, groups a set of CPF commands that define the power modes and power mode transitions that apply to the group defined by the preceding <code>set_power_mode_control_group</code> command.
<pre>get_parameter <i>parameter_name</i></pre>	Returns the value of a predefined parameter in the current design.

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<pre> identify_always_on_driver -pins <i>pin_list</i> [-no_propagation]</pre>	Specifies a list of pins in the design that must be driven by always-on buffer or inverter instances.
<pre> identify_power_logic -type isolation {-instances <i>instance_list</i> -module <i>name</i>}</pre>	Identifies any generic logic used for isolation that is instantiated in RTL or the gate-level netlist.
<pre> identify_secondary_domain -secondary_domain <i>domain</i> { -instances <i>instance_list</i> -cells <i>cell_list</i> } [-domain <i>power_domain</i> [-from <i>power_domain</i> -to <i>power_domain</i>]]</pre>	Identifies the secondary power domain for the selected instances with multiple power and ground pins. The primary power and ground nets of this secondary power domain are connected to the non-switchable power and ground pins of the identified instances.
<pre> include <i>file</i></pre>	Includes a CPF file or a Tcl file within a CPF file.
<pre> set_array_naming_style [<i>string</i>]</pre>	Specifies the format used to name the design objects in the netlist starting from multi-bit arrays in the RTL description.
<pre> set_cpf_version [<i>value</i>]</pre>	Specifies the version of the format.
<pre> set_design <i>module</i> [-ports <i>port_list</i>] [-honor_boundary_port_domain] [-parameters <i>parameter_value_list</i>]</pre>	Specifies the name of the module to which the power information in the CPF file applies.
<pre> set_equivalent_control_pins -master <i>pin</i> -pins <i>pin_list</i> { -domain <i>domain</i> -rules <i>rule_list</i> }</pre>	Specifies a list of pins that are equivalent with a master control pin. The master control pin is part of the definition of a shutoff condition, isolation condition or state retention condition. The referred condition can contain only the master control pin in its expression.
<pre> set_floating_ports <i>port_list</i></pre>	Specifies a list of ports of a macro cell that are not connected to any logic inside the macro cell.
<pre> set_hierarchy_separator [<i>character</i>]</pre>	Specifies the hierarchy delimiter character used in the CPF file.

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<pre>set_input_voltage_tolerance -ports <i>port_list</i> -bias [<i>float:</i>]<i>float</i></pre>	Specifies a list of input ports of a macro cell that can be driven by a signal with a higher voltage than the power supply voltage defined by the associated power domain of these ports without the need for level shifters.
<pre>set_instance [instance [-design <i>design</i> -model <i>macro_model</i>] [-port_mapping <i>port_mapping_list</i>] [-domain_mapping <i>domain_mapping_list</i>] [-parameter_mapping <i>parameter_mapping_list</i>]]</pre>	Changes the scope to the specified instance or links a previously defined CPF model to the specified instance.
<pre>set_macro_model <i>macro_cell</i></pre>	Indicates the start of the CPF content of a custom IP.
<pre>set_power_mode_control_group -name <i>group</i> { -domains <i>domain_list</i> -groups <i>group_list</i> -domains <i>domain_list</i> -groups <i>group_list</i>}</pre>	Groups a list of power domains and other power mode control groups.
<pre>set_power_target { -leakage <i>float</i> -dynamic <i>float</i> -leakage <i>float</i> -dynamic <i>float</i>}</pre>	Specifies the targets for the average leakage and dynamic power of the current design across all the power modes. All power targets must be specified in the units specified by the <code>set_power_unit</code> command.
<pre>set_power_unit [pW nW uW mW W]</pre>	Specifies the unit for all power values in the CPF file.
<pre>set_register_naming_style [<i>string</i>%s]</pre>	Specifies the format used to name flip-flops and latches in the netlist starting from the register names in the RTL description.
<pre>set_switching_activity { -all -pins <i>pin_list</i> -instances <i>instance_list</i> [-hierarchical]} { -probability <i>float</i> -toggle_rate <i>float</i> [-clock_pins <i>pin_list</i>] -toggle_percentage <i>float</i> } [-mode <i>mode</i>]</pre>	Specifies activity values (toggle rate and probability) for the specified pins.
<pre>set_time_unit [ns us ms]</pre>	Specifies the unit for all time values in the CPF file.
<pre>set_wire_feedthrough_ports <i>port_list</i></pre>	Specifies a list of input ports and output ports of a macro cell that are internally connected to each other by a physical wire only.

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<pre>update_isolation_rules -names rule_list { -location {from to} -within_hierarchy instance -cells cell_list -prefix string -open_source_pins_only}+</pre>	Appends the specified isolation rules with additional implementation information. At least one option besides <code>-name</code>, must be specified; several options may be updated together.
<pre>update_level_shifter_rules -names rule_list { -location {from to} -within_hierarchy instance -cells cell_list -prefix string}+</pre>	Appends the specified level shifter rule with implementation information.
<pre>update_nominal_condition -name condition -library_set library_set</pre>	Associates a library set with the specified operating condition.
<pre>update_power_domain -name domain { -primary_power_net net -primary_ground_net net -equivalent_power_nets list_of_power_nets -equivalent_ground_nets list_of_ground_nets -pmos_bias_net net -nmos_bias_net net -user_attributes string_list -transition_slope [float:]float -transition_latency { from_nom latency_list } -transition_cycles { from_nom cycle_list clock_pin } }+</pre>	Specifies implementation aspects of the specified power domain.
<pre>update_power_mode -name mode { -activity_file file -activity_file_weight weight -sdc_files sdc_file_list -setup_sdc_files sdc_file_list -hold_sdc_files sdc_file_list -peak_ir_drop_limit domain_voltage_list -average_ir_drop_limit domain_voltage_list -leakage_power_limit float -dynamic_power_limit float}+</pre>	Specifies the constraints for the power mode. At least one option besides <code>-name</code> must be specified; several options may be updated together
<pre>update_power_switch_rule -name string { -enable_condition_1 expression -enable_condition_2 expression -acknowledge_receiver_1 expression -acknowledge_receiver_2 expression -cells cell_list -gate_bias_net power_net -prefix string -peak_ir_drop_limit float -average_ir_drop_limit float }+</pre>	Appends the specified rules for power switch logic with implementation information.

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<pre>update_state_retention_rules -names <i>rule_list</i> { -cell_type <i>string</i> -cells <i>cell_list</i> -set_reset_control}+</pre>	Appends the specified rules for state retention logic with implementation information.
---	---

Commands to configure the Cell Library	
<pre>define_always_on_cell -cells <i>cell_list</i> [-library_set <i>library_set</i>] [{ -power_switchable <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i> -power_switchable <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i> } -power <i>LEF_power_pin</i> -ground <i>LEF_ground_pin</i>]</pre>	Identifies the library cells in the .lib files with more than one set of power and ground pins that can remain powered on even when the power domain they are instantiated in is powered down.
<pre>define_isolation_cell -cells <i>cell_list</i> [-library_set <i>library_set</i>] [-always_on_pins <i>pin_list</i>] [{ -power_switchable <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i> } -power <i>LEF_power_pin</i> -ground <i>LEF_ground_pin</i>] [-valid_location { from to on off}] {-enable <i>pin</i> -no_enable {high low hold}} [-non_dedicated]</pre>	Identifies the library cells in the .lib files that must be used as isolation cells.

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<pre>define_level_shifter_cell -cells <i>cell_list</i> [-library_set <i>library_set</i>] [-always_on_pins <i>pin_list</i>] { -input_voltage_range {<i>voltage</i> <i>voltage_range</i>} -output_voltage_range {<i>voltage</i> <i>voltage_range</i>} -ground_input_voltage_range {<i>voltage</i> <i>voltage_range</i>} -ground_output_voltage_range {<i>voltage</i> <i>voltage_range</i>} -input_voltage_range {<i>voltage</i> <i>voltage_range</i>} -output_voltage_range {<i>voltage</i> <i>voltage_range</i>} -ground_input_voltage_range {<i>voltage</i> <i>voltage_range</i>} -ground_output_voltage_range {<i>voltage</i> <i>voltage_range</i>} } [-direction {up down bidir}] [-input_power_pin <i>LEF_power_pin</i>] [-output_power_pin <i>LEF_power_pin</i>] [-input_ground_pin <i>LEF_ground_pin</i>] [-output_ground_pin <i>LEF_ground_pin</i>] [-ground <i>LEF_ground_pin</i>] [-power <i>LEF_power_pin</i>] [-enable <i>pin</i>] [-valid_location {to from either}]</pre>	Identifies the library cells in the .lib files that must be used as level shifter cells.
<pre>define_open_source_input_pin -cells <i>cell_list</i> -pin <i>pin</i> [-library_set <i>library_set</i>]</pre>	Specifies a list of cells that contain open source input pins.
<pre>define_power_clamp_cell -cells <i>cell_list</i> -data <i>pin</i> -power <i>pin</i> [-ground <i>pin_name</i>] [-library_set <i>library_set</i>]</pre>	Specifies a list of diode cells used for power clamp control.

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<pre>define_power_switch_cell -cells <i>cell_list</i> [-library_set <i>library_set</i>] -stage_1_enable <i>expression</i> [-stage_1_output <i>expression</i>] [-stage_2_enable <i>expression</i>] [-stage_2_output <i>expression</i>] -type {footer header} [-enable_pin_bias [<i>float</i>:]<i>float</i>] [-gate_bias_pin <i>LEF_power_pin</i>] [-power_switchable <i>LEF_power_pin</i> -power <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i> -ground <i>LEF_ground_pin</i>] [-stage_1_on_resistance <i>float</i>] [-stage_2_on_resistance <i>float</i>] [-stage_1_saturation_current <i>float</i>] [-stage_2_saturation_current <i>float</i>] [-leakage_current <i>float</i>]</pre>	Identifies the library cells in the .lib files that must be used as power switch cells.
<pre>define_related_power_pins -data_pins <i>pin_list</i> -cells <i>cell_list</i> [-library_set <i>library_set</i>] {-power <i>LEF_power_pin</i> -ground <i>LEF_ground_pin</i> -power <i>LEF_power_pin</i> -ground <i>LEF_ground_pin</i> }</pre>	Specifies the relationship between the power pins and data pins for cells that have more than one set of power and ground pins.
<pre>define_state_retention_cell -cells <i>cell_list</i> [-library_set <i>library_set</i>] [-cell_type <i>string</i>] [-always_on_pins <i>pin_list</i>] [-clock_pin <i>pin</i>] {-restore_function <i>expression</i> -save_function <i>expression</i> -restore_function <i>expression</i> -save_function <i>expression</i> } [-restore_check <i>expression</i>] [-save_check <i>expression</i>] [-always_on_components <i>component_list</i>] [{-power_switchable <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i> -power_switchable <i>LEF_power_pin</i> -ground_switchable <i>LEF_ground_pin</i>} -power <i>LEF_power_pin</i> -ground <i>LEF_ground_pin</i>]</pre>	Identifies the library cells in the .lib files that must be used as state retention cells.

Example of CPF Constructs

Example of CPF Constructs

```
#-----
set_design mod_B
create_power_domain -name PDX -default
create_state_retention_rule -name RETB1 -domain PDX
end_design mod_B

# CPF File of Top Design
#-----
set_design top
# Set up logic structure for all power domains
#-----
include IPB.cpf
create_power_domain -name PD1 -default

create_power_domain -name PD2 -instances {inst_A} \
-shutoff_condition {!pm_inst.pse_enable[0]} -base_domains PD1 \
-default_restore_edge {!pm_inst.pge_enable[0]}

create_power_domain -name PD3 -instances inst_C \
-shutoff_condition {!pm_inst.pse_enable[1]} -base_domains PD1

create_power_domain -name PD4 -instances inst_D \
-shutoff_condition {!pm_inst.pse_enable[2]} -base_domains PD1

# Define static behavior of all power domains and specify
# timing constraints
#-----
set_instance inst_B -domain_mapping { PDX PD2 }
create_nominal_condition -name high -voltage 1.2
create_nominal_condition -name medium -voltage 1.1
create_nominal_condition -name low -voltage 1.0

create_power_mode -name PM1 -domain_conditions \
{PD1@high PD2@medium PD3@high PD4@low}

update_power_mode -name PM1 -sdc_files ../SCRIPTS/cm1.sdc \
-activity_file ../SIM/top_1.tcf -activity_file_weight 1

create_power_mode -name PM2 -domain_conditions \
{PD1@high PD3@high PD4@low}

update_power_mode -name PM2 -sdc_files ../SCRIPTS/cm2.sdc
create_power_mode -name PM3 -domain_conditions {PD1@high PD4@low}
create_power_mode -name PM4 -domain_conditions {PD1@low}

# Set up required isolation and state retention rules for all domains
#-----
create_state_retention_rule -name sr1 -domain PD3 \
-restore_edge {!pm_inst.pge_enable[1]}

create_state_retention_rule -name sr2 -domain PD4 \
-restore_edge {!pm_inst.pge_enable[2]}
-isolation_condition {pm_inst.ice_enable[0]} -isolation_output high

create_isolation_rule -name ir2 -from PD3 \
-isolation_condition {pm_inst.ice_enable[1]}

create_isolation_rule -name ir3 -from PD4 \
-isolation_condition {pm_inst.ice_enable[2]}

create_level_shifter_rule -name lsr1 -to {PD1 PD3}
end_design
```

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